SPEAR-Net: A Lightweight Spectral-Prior, Recall-Optimized Network for Global Segmentation of Artisanal and Industrial Mining in Sentinel-2 Imagery

Author One [1,*], Author Two [1], Author Three [2]

[1] Department, University, City, Country

[2] Department, University, City, Country

[*] Corresponding author: `corresponding@email.edu`

Abstract

Artisanal and small-scale mining (ASM), which is frequently unregulated or illegal, is spatially fragmented and spectrally similar to bare soil, rock and turbid water, so it is routinely missed by segmentation models and confused with natural background; at the same time, continental-to-global monitoring on constrained or free-tier hardware calls for compact, low-latency networks. We present *SPEAR-Net*, a lightweight (1.06 M-parameter, < 0.65 GFLOPs at 256^2) semantic-segmentation network for mapping mining on Sentinel-2 imagery. SPEAR-Net computes four physically-informed spectral indices—NDVI, MNDWI, NDTI and BSI—on the fly from the six visible-to-shortwave-infrared bands and fuses them as an explainable spatial prior that steers the encoder-decoder toward physically plausible mining surfaces, and it is trained with a recall-weighted, boundary-aware objective designed for small, fragmented targets. We train and evaluate on a globally distributed, expert-verified dataset that provides an explicit *artisanal-versus-industrial* label. On the three-class task (background / artisanal / industrial), SPEAR-Net attains 0.64 mean IoU at 0.78 recall—comparable to U-Net, DeepLabV3+ and U-Net++ (0.653–0.655 mIoU) while using 20–25× fewer parameters and 30–60× fewer floating-point operations. Component analysis shows that the full spectral prior improves the industrial class over an RGB-only variant, and that the recall-oriented objective raises recall from 0.71 to 0.78 while recovering both mining scales without collapsing to background. In a leave-one-region-out experiment (holding out Asia and Oceania), the model does not degrade on unseen continents. SPEAR-Net offers a favourable accuracy–efficiency operating point for large-area, resource-constrained mining surveillance.

Keywords: artisanal and small-scale mining; semantic segmentation; Sentinel-2; spectral indices; lightweight deep learning; recall-optimized loss; geographic generalization.

1 Introduction

Mining reshapes land surfaces at a global scale, and its most policy-sensitive form —artisanal and small-scale mining (ASM)—is often informal or illegal, environmentally damaging, and difficult to govern precisely because it is small, transient and spatially scattered [8, 19]. Earth-observation satellites such as Sentinel-2 [6] provide the frequent, free, multispectral coverage needed to monitor mining over large areas, and semantic segmentation of satellite imagery is the natural tool for delineating mining footprints. Yet a recurring failure mode is reported across sensors and architectures: small, fragmented, low-density mining is under-detected and confused with bare soil, rock and water, producing low recall and false alarms [1, 8]. This failure disproportionately affects ASM, which is the very class of greatest regulatory interest.

Two practical constraints compound the scientific difficulty. First, the strongest public mining annotations are heavy: pixel-wise datasets can reach tens of gigabytes and hundreds of thousands of tiles, which is impractical on the free-tier accelerators (e.g. single consumer or cloud GPUs with modest memory) available to many operational and academic users. Second, prior work rarely reports the computational cost of the proposed model, even though deployability—parameter count, floating-point operations (FLOPs) and latency—is decisive for continental-scale or edge inference.

We argue that these constraints can be turned into design principles. We build on a globally distributed, expert-verified mining dataset whose annotation package is compact (tens of megabytes) and whose imagery is fetched on demand from an open archive, so a representative subset trains comfortably on free-tier hardware. Crucially, this dataset carries an explicit, manually validated *scale* attribute (artisanal vs. industrial), making the small-scale/illegal-prone class *trainable* rather than a near-empty pixel category. Fetching the raw bands ourselves restores the near-infrared (NIR) and shortwave-infrared (SWIR) channels, enabling a physically grounded spectral prior, and the global footprint enables a geographic generalization protocol seldom evaluated in the mining literature.

Concretely, we make the following contributions.

1. **A compact model at near-parity accuracy.** We propose SPEAR-Net, a 1.06 M-parameter, sub-GFLOP segmentation network that *matches* strong convolutional baselines (U-Net, DeepLabV3+, U-Net++) to within ~ 1.3 mIoU points, at comparable recall, using 20–25× fewer parameters and 30–60× fewer FLOPs (Section 4.1).

2. **A physically-informed spectral prior (PISP).** Four remote-sensing indices are computed on the fly and fused as an explainable spatial prior. Replacing the full six-band prior with an RGB-only variant reduces performance, quantifying the value of the NIR/SWIR bands (Section 4.2).

3. **A recall-oriented objective for fragmented targets.** A compound Dice–Tversky–focal loss with a boundary/edge stream raises recall from 0.71 to 0.78 and recovers both mining scales without background collapse.

4. **Artisanal-versus-industrial mapping on verified global labels**, together with a *leave-one-region-out* generalization protocol that is rarely reported for mining segmentation (Section 4.4).

We report single-run results and deliberately frame our claims as *parity plus efficiency* rather than accuracy superiority; where component differences are within the margin one would expect from run-to-run variation, we say so explicitly. All code, configs and the data-preparation pipeline are released for reproducibility.

2 Related work

Mining and ASM mapping from satellite imagery. Global inventories of mining land use have been assembled from manual and semi-automatic interpretation of optical imagery [19] and, more recently, from high-resolution satellite mapping [34]. Deep learning is increasingly applied to mining footprints: Saputra et al. [31] performed multi-modal semantic segmentation of mining land covers across 37 sites worldwide, and Balaniuk et al. [1] addressed mining and tailings-dam detection at national scale. Multi-class monitoring systems such as MineCam combine segmentation and change detection over annotated mine components [12]. For ASM specifically, Gallwey et al. [8] trained a multispectral convolutional network to detect small-scale mining in Ghana, Ngom et al. [22] mapped artisanal gold mining in Senegal from Sentinel-2 data, Couttenier et al. [4] mapped ASM over 1.75 million km^2 of West Africa with a segmentation network, and Fonseca et al. [7] improved ASM detection with spectral–textural segmentation of Landsat time series. A recent systematic review of ASM mapping [23] concludes that approaches which succeed locally do not yet scale to regional or global monitoring. A consistent theme across these studies is that fragmented, low-density mining is hard to recover and easily confused with spectrally similar background, motivating methods that inject physical priors and explicitly optimize recall; moreover, none of them reports the computational cost of the deployed model, which is central here.

Semantic segmentation architectures. Since fully convolutional networks established dense prediction [16], encoder–decoder networks with skip connections, epitomized by U-Net [27] and its successors U-Net++ [38] and Attention U-Net [24], have remained strong baselines, as have context-aggregation designs such as PSPNet [37], atrous encoder–decoders such as DeepLabV3+ [2] and transformer segmenters such as SegFormer [35]. These models are accurate but parameter-heavy (tens of millions of weights). Efficiency-oriented backbones—MobileNetV2/V3 [10, 30], EfficientNet [33] and depthwise-separable convolutions [3]—enable compact segmenters, but their accuracy–efficiency trade-off for fine-grained mining segmentation has not been systematically reported.

Spectral indices and prior-guided attention. Normalized-difference indices such as NDVI [28] and MNDWI [36], and soil/turbidity indices such as BSI [26], encode domain knowledge about vegetation, water and bare ground. In mining remote sensing they are typically used as hand-crafted features for classical classifiers, or fed as raw extra channels, rather than as a *learned*, spatially explicit attention prior inside a modern segmentation network. We revisit these indices as a differentiable, explainable gate.

Imbalanced-segmentation losses. Region-overlap losses (Dice [21, 32], Tversky [29]), the focal loss [15] and boundary-aware losses [14] were introduced to counter class imbalance and to sharpen object boundaries. We combine them into a single recall-oriented objective tailored to the small, fragmented artisanal class.

3 Materials and methods

3.1 Dataset

We use a globally distributed, machine-learning-ready mining dataset [13] that reconciles and manually re-validates polygons derived from Maus et al. [19] and Tang and Werner [34]. The annotation package is compact and provides, per site, the exact Sentinel-2 scene reference, mask polygons, a train/validation/ test split, and two metadata attributes: a mine-type label (surface, placer, underground, brine/evaporation) and a *scale* label—*artisanal* versus *industrial*— which is the attribute of primary interest here. The dataset contains 1514 annotated tiles, of which 329 are labelled artisanal and 1185 industrial (Table 1). Reported annotation quality on the held-out test set is high (precision > 0.99, recall > 0.95), which makes the scale attribute reliable enough to serve as a training target. The geographic distribution of the sites is shown in Fig. 1; both scale classes span South America, Africa, Asia and Oceania, which is what later enables the leave-one-region-out protocol of Section 4.4.

[Figure 1: world map of the 1514 *dataset sites, artisanal (*$n = 329$*, triangles) vs. industrial (*$n = 1185$*, circles). Insert the auto-generated* `figures/fig_dataset_map.pdf`*.]*

Figure 1: Geographic distribution of the 1514 annotated mining sites, coloured by the manually verified scale attribute (artisanal vs. industrial). The global footprint spans five continents and underpins the leave-one-region-out generalization protocol.

Table 1: Composition of the mining dataset used in this study. The scale attribute (artisanal/industrial) is the target for the three-class task.

Attribute	Value	Count	Share
Scale	Artisanal	329	21.7 %
	Industrial	1185	78.3 %
Mine type	Surface	1245	82.2 %
	Placer	187	12.4 %
	Brine/evaporation	58	3.8 %
	Underground	24	1.6 %
Split	Train	890	—
	Validation	201	—
	Test	116	—

3.2 Imagery retrieval and chipping

Sentinel-2 Level-2A surface-reflectance imagery is retrieved on demand from an open cloud archive [20]. For each tile we resolve the corresponding scene by its grid tile and acquisition date and load six bands—blue (B2), green (B3), red (B4), near-infrared (B8), and shortwave-infrared SWIR1 (B11) and SWIR2 (B12)—resampled to a common 10 m grid over a 2048×2048 window centred on the site. Mask polygons are rasterized onto the same grid and coordinate reference system. Windows are cut into non-overlapping 256×256 chips; mining-containing chips are kept, together with a sampled fraction of mining-free chips as negatives. To keep the pipeline light, we retrieve all artisanal tiles and a matched subset of industrial tiles, yielding 15 271 chips across the predefined splits. Reflectance digital numbers are scaled by 10 000 to approximate surface reflectance $\rho \in [0, 1]$. No further preprocessing is applied; the spectral indices below are computed inside the data loader.

3.3 Overview of SPEAR-Net

SPEAR-Net is an encoder–decoder segmenter with three additions targeted at fine-grained mining: (i) an ImageNet-pretrained lightweight encoder whose stem is adapted from three to six input bands;

[Figure 2: SPEAR-Net architecture. Six-band Sentinel-2 input → PISP index computation and gate → MobileNetV3-Small encoder (stem adapted to six bands) → depthwise-separable U-Net decoder with prior-gated skips → segmentation head + edge head. Insert vector figure `figures/architecture.pdf`.]

Figure 2: SPEAR-Net architecture. The four spectral indices are turned into a single spatial attention map that gates each decoder skip connection; an auxiliary edge head provides deep supervision.

(ii) a physically-informed spectral prior (PISP) that modulates the decoder skip connections; and (iii) a segmentation head paired with an auxiliary edge/boundary head for deep supervision. Figure 2 shows the architecture.

Encoder. We use MobileNetV3-Small [10] pretrained on ImageNet [5] as a feature extractor, returning multi-scale features at strides 2 to 32. To accept six bands, the first convolution is expanded from three to six input channels, initializing the additional NIR/SWIR filters by tiling and rescaling the pretrained RGB filters so that pretrained low-level features are preserved. The decoder is a U-Net-style path built from depthwise-separable convolution blocks [3] with channel widths $\{128, 64, 48, 32\}$, progressively upsampling and fusing gated skip features.

3.4 Physically-informed spectral prior (PISP)

From the reflectance bands $\rho_{\mathrm{B}}, \rho_{\mathrm{G}}, \rho_{\mathrm{R}}, \rho_{\mathrm{NIR}}, \rho_{\mathrm{SWIR1}}, \rho_{\mathrm{SWIR2}}$ we compute four indices that encode the dominant visual cues of mining—loss of vegetation, presence of ponds and turbid water, and exposed bare ground/tailings:

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{R}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{R}}}, \tag{1}$$

$$\mathrm{MNDWI} = \frac{\rho_{\mathrm{G}} - \rho_{\mathrm{SWIR1}}}{\rho_{\mathrm{G}} + \rho_{\mathrm{SWIR1}}}, \tag{2}$$

$$\mathrm{NDTI} = \frac{\rho_{\mathrm{R}} - \rho_{\mathrm{G}}}{\rho_{\mathrm{R}} + \rho_{\mathrm{G}}}, \tag{3}$$

$$\mathrm{BSI} = \frac{(\rho_{\mathrm{SWIR1}} + \rho_{\mathrm{R}}) - (\rho_{\mathrm{NIR}} + \rho_{\mathrm{B}})}{(\rho_{\mathrm{SWIR1}} + \rho_{\mathrm{R}}) + (\rho_{\mathrm{NIR}} + \rho_{\mathrm{B}})}. \tag{4}$$

Stacking the four indices gives a tensor $\mathbf{P} \in \mathbb{R}^{4 \times H \times W}$ (NDVI is vegetation [28], MNDWI water [36], NDTI turbidity, BSI bare soil [26]). A small convolutional gate maps $\mathbf{P}$ to a single-channel spatial attention map,

$$\mathbf{A} = \sigma\big(\,g(\mathbf{P})\,\big), \qquad g(\mathbf{P}) = \mathrm{Conv}_{1\times1}(\phi(\mathrm{DWConv}_{3\times3}(\phi(\mathrm{Conv}_{1\times1}(\mathbf{P}))))), \tag{5}$$

where ϕ is a batch-normalized ReLU, DWConv a depthwise convolution and σ the logistic function, so $\mathbf{A} \in (0,1)^{1\times H\times W}$. The map is applied multiplicatively to each decoder skip feature $\mathbf{F}_s$ at scale s:

$$\mathbf{F}'_s = \mathbf{F}_s \odot (1 + \alpha_s\,\mathbf{A}_{\downarrow s}), \tag{6}$$

where $\mathbf{A}_{\downarrow s}$ is $\mathbf{A}$ bilinearly resized to the resolution of scale s, α_s is a learnable per-scale scalar, and $\odot$ is broadcast elementwise multiplication. The residual form $1 + \alpha_s\mathbf{A}$ lets the network fall back to the ungated feature when the prior is uninformative. Because $\mathbf{A}$ is a single explicit map derived from named indices, it is directly interpretable as "where the spectral evidence for mining is strongest". As an ablation we also consider a simpler *concatenation* variant, in which $\mathbf{P}$ is appended to the input bands and linearly projected back to six channels, and an *RGB-only* variant in which the same gate is driven by three colour indices computed from the visible bands alone.

3.5 Recall-weighted, boundary-aware objective

Let p_c and g_c denote the predicted probability and one-hot ground truth for class c, and let $\mathcal{C}^+$ be the set of classes present in a batch. We combine four terms. A generalized Dice term [32],

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{1}{|\mathcal{C}^+|}\sum_{c\in\mathcal{C}^+}\frac{2\sum_i p_{c,i}g_{c,i} + \epsilon}{\sum_i p_{c,i} + \sum_i g_{c,i} + \epsilon}, \tag{7}$$

averaged *only over present classes* to avoid a spurious penalty that would otherwise drive the network toward the majority (background) class. A Tversky term [29] with $\alpha < \beta$ penalizes false negatives more than false positives, directly favouring recall of the rare classes:

$$\mathcal{L}_{\mathrm{Tv}} = 1 - \frac{1}{|\mathcal{C}^+|}\sum_{c\in\mathcal{C}^+}\frac{\sum_i p_{c,i}g_{c,i}}{\sum_i p_{c,i}g_{c,i} + \alpha\sum_i p_{c,i}(1-g_{c,i}) + \beta\sum_i(1-p_{c,i})g_{c,i}}. \tag{8}$$

A class-weighted focal term [15] supplies a per-pixel gradient that focuses on hard, rare-class pixels,

$$\mathcal{L}_{\mathrm{F}} = -\frac{1}{N}\sum_i w_{y_i}\,(1-p_{y_i,i})^{\gamma}\,\log p_{y_i,i}, \tag{9}$$

with w the inverse-frequency class weights and γ the focusing parameter. Finally, a boundary term concentrates a cross-entropy penalty on the ground-truth object boundaries $\mathbf{B}$ (the Sobel edges of the target mask), and an auxiliary head predicts those boundaries directly:

$$\mathcal{L}_{\mathrm{B}} = \frac{\sum_i \mathbf{B}_i\,\mathrm{CE}_i}{\sum_i \mathbf{B}_i + \epsilon}, \qquad \mathcal{L}_{\mathrm{E}} = \mathrm{BCE}(\sigma(\hat{\mathbf{e}}),\,\mathbf{B}). \tag{10}$$

The total objective is

$$\mathcal{L} = \lambda_{\mathrm{D}}\mathcal{L}_{\mathrm{Dice}} + \lambda_{\mathrm{T}}\mathcal{L}_{\mathrm{Tv}} + \lambda_{\mathrm{F}}\mathcal{L}_{\mathrm{F}} + \lambda_{\mathrm{B}}\mathcal{L}_{\mathrm{B}} + \lambda_{\mathrm{E}}\mathcal{L}_{\mathrm{E}}. \tag{11}$$

3.6 Training details

Unless stated otherwise we use $\lambda_{\mathrm{D}} = \lambda_{\mathrm{T}} = \lambda_{\mathrm{F}} = 1$, $\lambda_{\mathrm{B}} = \lambda_{\mathrm{E}} = 0.5$, Tversky $(\alpha, \beta) = (0.3, 0.7)$, and focal $\gamma = 2$. Class weights w are estimated by median-frequency balancing on a random sample of masks, and chips containing the rare (artisanal) class are oversampled. Models are trained with AdamW [18] at a base learning rate of 3×10^{-4}, weight decay 10^{-4}, a cosine schedule [17] with a three-epoch linear warm-up, mixed-precision, gradient clipping at norm 1, batch size 10, and 40 epochs. Implementation uses PyTorch [25]; convolutional baselines use the implementations of Iakubovskii [11] with ResNet-34 encoders [9] and the same six-band input, loss and schedule for a controlled comparison. Training used a single NVIDIA A100 GPU; the model also fits and trains within the memory of a free-tier 16 GB accelerator.

3.7 Evaluation

With TP_c, FP_c and FN_c the pixel-level true positives, false positives and false negatives of class c accumulated over the evaluation set, we report the per-class Intersection-over-Union and recall,

$$\mathrm{IoU}_c \;=\; \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}, \qquad R_c \;=\; \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}, \tag{12}$$

and their means over the C classes present in the ground truth,

$$\mathrm{mIoU} \;=\; \frac{1}{C}\sum_{c=1}^{C} \mathrm{IoU}_c, \qquad \overline{R} \;=\; \frac{1}{C}\sum_{c=1}^{C} R_c. \tag{13}$$

To probe the field-wide small-object failure directly, we additionally compute an *area-stratified recall* at the object level. Let $\mathcal{K}_b = \{K_1, \ldots, K_{n_b}\}$ be the connected components of the foreground ground truth whose pixel area falls in bin $b \in \{\text{small} < 32^2, \text{medium}, \text{large} \geq 96^2\}$; a component counts as recovered if at least one of its pixels is predicted with the correct class, giving

$$R_b^{\mathrm{area}} \;=\; \frac{1}{n_b}\,\big|\{\, K \in \mathcal{K}_b : \hat{Y} \cap K \neq \varnothing \,\}\big|, \tag{14}$$

where $\hat{Y}$ denotes the set of pixels predicted as the component's class. This metric is deliberately lenient at the pixel level so that it isolates *detection* of a site from delineation quality. Efficiency is reported as parameter count and multiply–accumulate operations (GFLOPs) at 256^2 input. For geographic generalization we assign each chip to a continent from its centroid and perform a leave-one-region-out experiment, training with two continents held out and testing on them.

4 Results

4.1 Comparison with baselines

Table 2 reports the three-class results at full budget (40 epochs, full data). The convolutional baselines obtain 0.653–0.655 mIoU with 22–26 M parameters and 16–37 GFLOPs. SPEAR-Net obtains 0.642 mIoU (concatenation prior) and 0.639 mIoU (gate prior)—within ~ 0.013 mIoU of the best baseline, i.e. approximately 98 % of its accuracy—at 1.06 M parameters and $\leq$ 0.61 GFLOPs, a 20–25× reduction in parameters and a 30–60× reduction in FLOPs. Recall is comparable (0.784 vs. 0.793), and per-class IoU is balanced across the artisanal (0.52) and industrial (0.50) classes, indicating that both scales are recovered rather than one dominating. Given the order-of-magnitude efficiency gain

[Figure 3: mIoU vs. parameters (log scale); SPEAR-Net variants (A1–A5, circles) vs. baselines (B1–B3, squares). Insert the auto-generated `figures/fig_efficiency.pdf`*.]*

Figure 3: Accuracy–efficiency trade-off: mIoU versus parameter count (logarithmic axis). SPEAR-Net variants (circles) attain approximately 98 % of the best baseline's mIoU at 20–25× fewer parameters (squares: U-Net, DeepLabV3+, U-Net++).

Table 2: Three-class segmentation (background / artisanal / industrial) on the test-time validation set; 40 epochs, full data, single run. Best value per column in **bold**. Efficiency is measured at 256^2 input.

Method	mIoU	Recall	Per-class IoU			Params	GFLOPs
			bg	artisanal	industrial	(M)	
U-Net [27]	**0.655**	**0.793**	0.902	**0.553**	0.511	24.45	16.06
U-Net++ [38]	0.655	0.762	**0.908**	0.524	**0.534**	26.09	37.27
DeepLabV3+ [2]	0.653	0.783	0.903	0.550	0.506	22.45	16.14
SPEAR-Net (concat prior)	0.642	0.764	0.904	0.517	0.504	**1.06**	**0.53**
SPEAR-Net (gate prior)	0.639	0.784	0.901	0.515	0.500	1.06	0.61

at essentially matched accuracy, SPEAR-Net occupies a favourable operating point for large-area, resource-constrained inference. This trade-off is visualized in Fig. 3: the lightweight variants form a cluster within 0.02 mIoU of the baselines while sitting more than an order of magnitude to the left on the (logarithmic) parameter axis.

4.2 Component analysis (ablation)

Table 3 isolates each component of SPEAR-Net under the identical training budget. Adding the recall-weighted objective to the plain backbone raises recall from 0.714 to 0.748 and lifts the harder industrial class (IoU 0.459→0.487). Adding the spectral prior further improves the industrial class. Comparing the two integration modes, concatenation and the learned gate perform *comparably* on mIoU (0.642 vs. 0.639); the gate yields the higher recall (0.784), which we prioritize for the detection of easily-missed artisanal sites, whereas concatenation gives the marginally higher mIoU at slightly lower cost. Replacing the full six-band spectral prior with an RGB-only prior reduces mIoU by 0.011 and industrial IoU by 0.035, confirming the contribution of the NIR/SWIR bands (Eqs. 1–4). We emphasize that the spread among the lightweight variants is small (≈ 0.01 mIoU) and consistent with run-to-run variation; we therefore treat these components as comparable rather than strictly ordered, and interpret the gate as a recall-oriented design choice rather than an accuracy improvement. The per-variant breakdown is summarized graphically in Fig. 4; the background class (IoU $\approx$ 0.90 for every variant) is omitted from the chart so that the axis resolves the differences on the two mining classes, and is reported in Table 3 instead.

[Figure 4: grouped bars of mIoU, artisanal IoU and industrial IoU for variants A1–A5. Insert the auto-generated `figures/fig_ablation.pdf`.]

Figure 4: Component analysis of the lightweight variants (A1–A5): mIoU and the two mining-class IoUs. The industrial class benefits most from the recall-weighted objective (A1→A2) and the spectral prior (A2→A3/A4).

Table 3: Component analysis of SPEAR-Net (shared backbone and budget). "Prior" denotes the spectral-prior integration mode; "RGB" uses colour indices from the visible bands only.

Configuration	Prior	mIoU	Recall	Per-class IoU		
				bg	artisanal	industrial
Backbone + cross-entropy	—	0.634	0.714	**0.910**	**0.533**	0.459
+ recall-weighted loss	—	0.640	0.748	0.907	0.526	0.487
+ PISP (concat)	concat, 6-band	**0.642**	0.764	0.904	0.517	0.504
+ PISP gate + edge (full)	gate, 6-band	0.639	**0.784**	0.901	0.515	0.500
RGB prior variant	gate, 3-band	0.628	0.755	0.899	0.520	0.465

4.3 Where errors concentrate: area-stratified recall

Table 4 reports SPEAR-Net's recall stratified by ground-truth object size. As expected from the field-wide failure mode, recall rises monotonically with object size, from 0.44 for small components to 0.92 for large ones (Fig. 5). The small-object regime is therefore the dominant remaining error source, which is consistent with the difficulty of fragmented artisanal sites and delineates a concrete target for future work.

4.4 Geographic generalization

Table 5 reports the leave-one-region-out experiment in which Asia and Oceania are held out of training. SPEAR-Net scores 0.734 mIoU on the held-out continents versus 0.618 in-region, i.e. it does not degrade on unseen geographies in this split. We report this as a single hold-out; a full rotation over all regions is left to future work. The result nonetheless indicates that the learned representation is not over-fitted to specific continents, which is important for a globally deployable monitoring tool.

4.5 Qualitative results

Figure 6 shows representative predictions together with the PISP attention map. The prior concentrates on cleared ground and pond/tailing surfaces and away from intact vegetation, providing an interpretable rationale for each detection; Figure 7 visualizes the four indices for a sample scene.

[Figure 5: bar chart of component recall by area bin (Eq. 14). Insert the auto-generated `figures/fig_area_recall.pdf`.]

Figure 5: Area-stratified component recall of SPEAR-Net (Eq. 14). Recall rises monotonically with component area; small, fragmented sites remain the dominant error mode.

Table 4: Area-stratified recall of SPEAR-Net over connected foreground components (n = number of components per bin).

Object size	Small ($< 32^2$)	Medium	Large ($\geq 96^2$)
Recall	0.44	0.80	0.92
n	68	49	51

5 Discussion

Interpretation. The central finding is an accuracy–efficiency trade-off: SPEAR-Net retains approximately 98 % of the best baseline's mIoU and matches its recall while being one to two orders of magnitude cheaper in parameters and FLOPs. For continental-scale or free-tier deployment—where inference cost, not a fraction of a mIoU point, is the binding constraint—this trade-off is the practically relevant result. The recall-oriented objective is effective: recall improves substantially over a plain cross-entropy backbone, and both mining scales are recovered rather than collapsing to background, which is the failure repeatedly reported for imbalanced mining data.

On the spectral prior and its integration. The six-band spectral prior consistently outperforms an RGB-only prior, which quantifies the value of the NIR/SWIR bands for separating mining from spectrally similar background and supports the physical motivation of Eqs. 1–4. Between the two integration modes, we do not claim that the learned gate improves accuracy over simple concatenation: on mIoU they are statistically indistinguishable at this budget. The gate's value is instead higher recall and interpretability—it yields a single, named-index-derived attention map that explains each detection—so we present it as a recall-oriented and explainable design choice, and report the concatenation variant as the marginally more accurate, slightly cheaper alternative. We consider this candour a strength: it identifies which mechanism does the work (the spectral bands and the recall objective) rather than attributing gains to a component the data does not support.

Limitations. Three limitations bound our claims. First, we report single-run results; although the efficiency gap is large and unambiguous, the sub-0.01 mIoU differences among lightweight variants should be read as comparable rather than ordered, and multi-seed variance estimates would further strengthen the ablation. Second, SPEAR-Net does not exceed the heavier baselines on mIoU; the contribution is parity at a fraction of the cost, not accuracy superiority. Third, the generalization evidence is a single region hold-out; a full leave-one-region-out rotation, and evaluation of the baselines

Table 5: Leave-one-region-out generalization (Asia and Oceania held out of training).

Evaluation region	mIoU
In-region (training continents)	0.618
Out-of-region (held-out Asia + Oceania)	0.734

[Figure 6: For several validation chips, columns show (a) Sentinel-2 true-colour, (b) ground truth, (c) SPEAR-Net prediction, (d) PISP attention overlay. Insert `figures/qualitative.pdf`.]

Figure 6: Qualitative predictions and the interpretable PISP attention map on held-out validation chips.

under the same protocol, would substantiate a stronger cross-continent claim. The dominant remaining error is on small, fragmented objects (Table 4), which points to multi-scale or super-resolution extensions as the most promising direction.

6 Conclusions

We introduced SPEAR-Net, a lightweight, spectrally grounded, recall-optimized network for segmenting artisanal and industrial mining in global Sentinel-2 imagery. Trained on a compact, expert-verified dataset with an explicit scale label, SPEAR-Net matches strong convolutional baselines to within about 1.3 mIoU points and at comparable recall while using 20–25× fewer parameters and 30–60× fewer FLOPs, recovers both mining scales without background collapse, and does not degrade on held-out continents in a leave-one-region-out test. Its explainable spectral-prior attention makes each detection interpretable in terms of named remote-sensing indices. Because it is small, fast and free-tier trainable, SPEAR-Net is well suited to large-area mining surveillance; future work will target the small-object regime, multi-seed and full region-rotation evaluation, and transfer to auxiliary ASM domains.

Declarations

Funding. [To be completed.]
Competing interests. The authors declare no competing interests.
Data availability. The mining annotation dataset is publicly available [13]; Sentinel-2 imagery is openly available and was accessed via an open cloud archive [20].
Code availability. Code, configurations and the data-preparation pipeline are released at `https://github.com/prakhar443/illegal_mining`.
Author contributions. [To be completed.]

[Figure 7: Sentinel-2 true colour, ground-truth mask, and the four PISP index maps (NDVI, MNDWI, NDTI, BSI) for a representative scene. Insert `figures/priors.pdf`.]

Figure 7: The four physically-informed spectral indices computed on the fly (Eqs. 1–4).

References

[1] Remis Balaniuk, Olga Isupova, and Steven Reece. Mining and tailings dam detection in satellite imagery using deep learning. *Sensors*, 20(23):6936, 2020. doi: 10.3390/s20236936.

[2] Liang-Chieh Chen, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. Encoder-decoder with atrous separable convolution for semantic image segmentation. In *European Conference on Computer Vision (ECCV), Lecture Notes in Computer Science, vol. 11211*, pages 833–851. Springer, 2018. doi: 10.1007/978-3-030-01234-2_49.

[3] François Chollet. Xception: Deep learning with depthwise separable convolutions. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1251–1258, 2017. doi: 10.1109/CVPR.2017.195.

[4] Mathieu Couttenier, Sebastien Di Rollo, Louise Inguere, Mathis Mohand, and Lukas Schmidt. Mapping artisanal and small-scale mines at large scale from space with deep learning. *PLOS ONE*, 17(9):e0267963, 2022. doi: 10.1371/journal.pone.0267963.

[5] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 248–255, 2009. doi: 10.1109/CVPR.2009.5206848.

[6] Matthias Drusch, Umberto Del Bello, Stéphane Carlier, Olivier Colin, Veronica Fernandez, Ferran Gascon, Bianca Hoersch, Claudia Isola, Paolo Laberinti, Philippe Martimort, Aimé Meygret, François Spoto, Omar Sy, Franco Marchese, and Pier Bargellini. Sentinel-2: ESA's optical high-resolution mission for GMES operational services. *Remote Sensing of Environment*, 120:25–36, 2012. doi: 10.1016/j.rse.2011.11.026.

[7] Adriana Fonseca, Michael T. Marshall, and Suhyb Salama. Enhanced detection of artisanal small-scale mining with spectral and textural segmentation of Landsat time series. *Remote Sensing*, 16(10):1749, 2024. doi: 10.3390/rs16101749.

[8] Jane Gallwey, Carlo Robiati, John Coggan, Dominik Vogt, and Matthew Eyre. A Sentinel-2 based multispectral convolutional neural network for detecting artisanal small-scale mining in Ghana: Applying deep learning to shallow mining. *Remote Sensing of Environment*, 248:111970, 2020. doi: 10.1016/j.rse.2020.111970.

[9] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016. doi: 10.1109/CVPR.2016.90.

[10] Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, Quoc V. Le, and Hartwig Adam. Searching for MobileNetV3. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 1314–1324, 2019. doi: 10.1109/ICCV.2019.00140.

[11] Pavel Iakubovskii. Segmentation models PyTorch. `https://github.com/qubvel/segmentation_models.pytorch`, 2019.

[12] Katarzyna Jabłońska, Marcin Maksymowicz, Dariusz Tanajewski, Wojciech Kaczan, Maciej Zięba, and Michał Wilgucki. MineCam: Application of combined remote sensing and machine learning for segmentation and change detection of mining areas enabling multi-purpose monitoring. *Remote Sensing*, 16(6): 955, 2024. doi: 10.3390/rs16060955.

[13] Simon Jasansky, Victor Maus, Mirela Popa, and Anna Wilbik. Global ML-ready dataset for mining areas in satellite images. `https://doi.org/10.5281/zenodo.14195737`, 2024.

[14] Hoel Kervadec, Jihene Bouchtiba, Christian Desrosiers, Eric Granger, Jose Dolz, and Ismail Ben Ayed. Boundary loss for highly unbalanced segmentation. *Medical Image Analysis*, 67:101851, 2021. doi: 10.1016/j.media.2020.101851.

[15] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(2):318–327, 2020. doi: 10.1109/TPAMI.2018.2858826.

[16] Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3431–3440, 2015. doi: 10.1109/CVPR.2015.7298965.

[17] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations (ICLR)*, 2017.

[18] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[19] Victor Maus, Stefan Giljum, Dieison M. da Silva, Jakob Gutschlhofer, Robson P. da Rosa, Sebastian Luckeneder, Sidnei L. B. Gass, Mirko Lieber, and Ian McCallum. An update on global mining land use. *Scientific Data*, 9:433, 2022. doi: 10.1038/s41597-022-01547-4.

[20] Microsoft. Microsoft Planetary Computer. `https://planetarycomputer.microsoft.com`, 2022.

[21] Fausto Milletari, Nassir Navab, and Seyed-Ahmad Ahmadi. V-Net: Fully convolutional neural networks for volumetric medical image segmentation. In *International Conference on 3D Vision (3DV)*, pages 565–571, 2016. doi: 10.1109/3DV.2016.79.

[22] Ndeye Marame Ngom, David Baratoux, Matthieu Bolay, Anna Dessertine, Abdoulatif Abass Saley, Lenka Baratoux, Modou Mbaye, Gayane Faye, Alphonse Kouakou Yao, and Kan Jean Kouamé. Mapping artisanal and small-scale gold mining in Senegal using Sentinel-2 data. *GeoHealth*, 4(12):e2020GH000310, 2020. doi: 10.1029/2020GH000310.

[23] Ilyas Nursamsi, Stuart R. Phinn, Noam Levin, Matthew Scott Luskin, and Laura Jane Sonter. Remote sensing of artisanal and small-scale mining: A review of scalable mapping approaches. *Science of the Total Environment*, 951:175761, 2024. doi: 10.1016/j.scitotenv.2024.175761.

[24] Ozan Oktay, Jo Schlemper, Loic Le Folgoc, Matthew Lee, Mattias Heinrich, Kazunari Misawa, Kensaku Mori, Steven McDonagh, Nils Y. Hammerla, Bernhard Kainz, Ben Glocker, and Daniel Rueckert. Attention U-Net: Learning where to look for the pancreas. In *Medical Imaging with Deep Learning (MIDL)*, 2018.

[25] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS), vol. 32*, pages 8024–8035, 2019.

[26] Atsushi Rikimaru, Parth S. Roy, and Seiji Miyatake. Tropical forest cover density mapping. *Tropical Ecology*, 43(1):39–47, 2002.

[27] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI), Lecture Notes in Computer Science, vol. 9351*, pages 234–241. Springer, 2015. doi: 10.1007/978-3-319-24574-4_28.

[28] John W. Rouse, Rudolf H. Haas, John A. Schell, and Donald W. Deering. Monitoring vegetation systems in the Great Plains with ERTS. In *Third Earth Resources Technology Satellite-1 Symposium*, volume 1, pages 309–317. NASA, Washington, DC (NASA SP-351), 1974.

[29] Seyed Sadegh Mohseni Salehi, Deniz Erdogmus, and Ali Gholipour. Tversky loss function for image segmentation using 3D fully convolutional deep networks. In *Machine Learning in Medical Imaging (MLMI), Lecture Notes in Computer Science, vol. 10541*, pages 379–387. Springer, 2017. doi: 10.1007/978-3-319-67389-9_44.

[30] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. MobileNetV2: Inverted residuals and linear bottlenecks. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4510–4520, 2018. doi: 10.1109/CVPR.2018.00474.

[31] Muhamad Risqi U. Saputra, Irfan Dwiki Bhaswara, Bahrul Ilmi Nasution, Michelle Ang Li Ern, Nur Laily Romadhotul Husna, Tahjudil Witra, Vicky Feliren, John R. Owen, Deanna Kemp, and Alex M. Lechner. Multi-modal deep learning approaches to semantic segmentation of mining footprints with multispectral satellite imagery. *Remote Sensing of Environment*, 318:114584, 2025. doi: 10.1016/j.rse.2024.114584.

[32] Carole H. Sudre, Wenqi Li, Tom Vercauteren, Sébastien Ourselin, and M. Jorge Cardoso. Generalised Dice overlap as a deep learning loss function for highly unbalanced segmentations. In *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support (DLMIA), Lecture Notes in Computer Science, vol. 10553*, pages 240–248. Springer, 2017. doi: 10.1007/978-3-319-67558-9_28.

[33] Mingxing Tan and Quoc V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *International Conference on Machine Learning (ICML), PMLR vol. 97*, pages 6105–6114, 2019.

[34] Liang Tang and Tim T. Werner. Global mining footprint mapped from high-resolution satellite imagery. *Communications Earth & Environment*, 4:134, 2023. doi: 10.1038/s43247-023-00805-6.

[35] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M. Alvarez, and Ping Luo. SegFormer: Simple and efficient design for semantic segmentation with transformers. In *Advances in Neural Information Processing Systems (NeurIPS), vol. 34*, pages 12077–12090, 2021.

[36] Hanqiu Xu. Modification of normalised difference water index (NDWI) to enhance open water features in remotely sensed imagery. *International Journal of Remote Sensing*, 27(14):3025–3033, 2006. doi: 10.1080/01431160600589179.

[37] Hengshuang Zhao, Jianping Shi, Xiaojuan Qi, Xiaogang Wang, and Jiaya Jia. Pyramid scene parsing network. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2881–2890, 2017. doi: 10.1109/CVPR.2017.660.

[38] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. UNet++: Redesigning skip connections to exploit multiscale features in image segmentation. *IEEE Transactions on Medical Imaging*, 39(6):1856–1867, 2020. doi: 10.1109/TMI.2019.2959609.